Shannon (1948): capacity of a discrete noiseless channel

Notes from Carlos' lessons

Source lesson: 2026-08-14 (Shannon's paper, part 1). See also `~/repos/books/entropy.pdf` (the paper), Part I, Section 1, "The discrete noiseless channel".

Setting and definition

A channel transmits sequences of symbols $S_1, \dots, S_n$ with durations $t_1, \dots, t_n$ (for example telegraph dots, dashes and spaces). Let $N(T)$ be the number of distinct allowed signals of duration T.

Definition 1 (Capacity (board, with units))**.**

$$C = \lim_{T \to \infty} \frac{\log N(T)}{T}.$$

With $\log_2$ the unit is **bits** (for example, if there are 32 messages then $\log_2 32 = 5$ bits), and C is measured in bits per second (the board's "bits/second").

Case 1: free choice of symbols

Recurrence. Condition on the last symbol: a signal of duration T ends with some S_i (duration t_i), preceded by an arbitrary signal of duration $T - t_i$:

$$N(T) = N(T - t_1) + N(T - t_2) + \cdots + N(T - t_n).$$

This is a linear recurrence with constant coefficients, so its solutions are combinations $c_1 \lambda_1^T + \cdots + c_n \lambda_n^T$ where the λ_i are roots of the characteristic equation. For large T the term with the largest real root λ_0 dominates: $N(T) = \Theta(\lambda_0^T)$ (the board's "$\Theta(\lambda^t)$").

Substituting $N(T) = \lambda^T$ into the recurrence and dividing by λ^T gives

$$1 = \lambda^{-t_1} + \cdots + \lambda^{-t_n}.$$

So λ_0 is the largest real root of this equation, and

$$C = \lim \frac{\log N(T)}{T} = \log \lambda_0.$$

Example 2 (Shannon's telegraph example)**.** Four symbols: dot = 2, dash = 4, letter space = 3, word space = 6 time units, with two spaces never adjacent. The effective "symbols" are dot, dash, dot+letter space, dash+letter space, dot+word space, dash+word space, of durations $2, 4, 5, 7, 8, 10$. The equation is

$$\lambda^{-2} + \lambda^{-4} + \lambda^{-5} + \lambda^{-7} + \lambda^{-8} + \lambda^{-10} = 1,$$

with largest root $\lambda_0 \approx 1.4529$, so $C = \log_2 \lambda_0 \approx 0.539$ bits per time unit (computed locally).

Case 2: a finite-state channel (the DFA on the board)

Now only some symbol sequences are allowed. Model this as a finite automaton with states $1, \ldots, p$. In state i the allowed symbols are indexed by s; the s-th one takes duration $b_{ij}^{(s)}$ and leads to state j. (On the board: the DFA with two states and the durations $b_{12}^{(1)}, b_{12}^{(2)}, \ldots.$)

Let $N_j(T)$ be the number of signals of duration T that **end in state** j. Conditioning on the last transition,

$$N_j(T) = \sum_{i=1}^{p} \sum_{s} N_i(T - b_{ij}^{(s)}), \qquad j = 1, \ldots, p,$$

and $N(T) = N_1(T) + \cdots + N_p(T)$.

Ansatz. For large T, $N_j(T) \sim k_j \lambda^T$ with the *same* growth rate λ for every j (the board: "λ independent of j"). Substituting and dividing by λ^T,

$$k_j = \sum_{i} \sum_{s} k_i \lambda^{-b_{ij}^{(s)}}, \qquad j = 1, \ldots, p.$$

In matrix form, with $M(\lambda)_{ij} = \sum_s \lambda^{-b_{ij}^{(s)}}$ (the board's $\left[\sum_s \lambda^{-b_{ij}^{(s)}}\right]_{i,j}$), this reads $\vec{k} = M(\lambda)^{\mathsf{T}} \vec{k}$, i.e. $(M(\lambda)^{\mathsf{T}} - I_p)\vec{k} = \vec{0}$. A **non-trivial** solution $\vec{k} \neq 0$ exists iff the matrix is **not invertible** (the board's remark: $M\vec{v} = \vec{0}$ has a non-trivial solution iff M is not invertible), that is,

$$\det\Big(\Big[\sum_{s} \lambda^{-b_{ij}^{(s)}}\Big]_{ij} - I_p\Big) = 0$$

(transposing does not change the determinant). The growth rate λ_0 is the **largest real root** of this determinant equation, and then $N(T) = \Theta(\lambda_0^T)$, so

$$\boxed{C = \log \lambda_0}.$$

This is Shannon's Theorem 1: *"the channel capacity C is $\log W$, where W is the largest real root of the determinant equation $\left|\sum_s W^{-b_{ij}^{(s)}} - \delta_{ij}\right| = 0$"* ($\delta_{ij}$ is the Kronecker delta, i.e. I_p). Case 1 is the special case of a single state.

Notes and gaps

- **What is not proved on the board:** that $N_j(T) \sim k_j \lambda^T$ at all, and that the dominant root is real and the right one. The rigorous argument is Perron–Frobenius for the non-negative matrix $M(\lambda)$ (for an irreducible channel). The board, like Shannon, uses the ansatz.

- Shannon's own treatment is in Part I, Section 1 of the paper. Confirmed by search: he defines $C = \lim_{T\to\infty} \frac{\log N(T)}{T}$, derives $N(T) = N(T-t_1) + \cdots + N(T-t_n)$, and states that for large T the count is dominated by X_0^T, where X_0 is the largest real root of $X^{-t_1} + \cdots + X^{-t_n} = 1$, giving $C = \log X_0$.
- Sources: Shannon 1948 (UVM copy), Yale copy, reading notes.
- The top of the board (L_1][L_2, with $2^{L_1+L_2}$ crossed out) is a scratch comparison of concatenated messages: lengths add under concatenation but the *count* of messages multiplies, which is why one takes $\log N$ to get something additive ($\log_2$ of a product is the sum of the logs).